Measurement Map of the Enhancement Paper

Companion to `enhancement/ENHANCEMENT.tex`, 28 October 2026

1 Scope

The map names every object of the model of the enhancement paper once, and the manuscript uses these names and no others. Seven names are those of the parent bundle's terminology table, bank, supervisor, insurer, investors, discount rate, dividend barrier and reference. The other 22 were agreed with the author for this paper on 28 October 2026. Inputs are what the model takes as given, internal objects are built from the inputs inside the model, and outputs are what the rows predict. Directly observed is Yes when a record of the world gives the object without any computation, which is when the last column begins with Recorded, and No otherwise. The other status words are Elicited, Computed, Assumed, Never observed and A device. Every computed object names its recovery formula and the Lean theorem that carries it, in `lean/mathlib/`. The data are the FDIC Call Report files of `common_data/`, and the facts are those fixed in `comparison/PREREG_comparison.md`. In the assumptions, the letter d is the differential, t is time and i indexes banks, and none of them is an object of the model.

1.1 Inputs

Table 1: Inputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
k	**capital ratio**	A real number above zero.	Total capital, tier 1 plus tier 2, over risk-weighted assets, for each bank and quarter.	No	Computed as `(RBCT1J+RBCT2)/RWAJT` from the Call Report, fields defined in the record lit/fdic_2026. The ratio is free of the unit of account (`EnhancementMap.capital_ratio_units`).
$\underline{k}$	**regulatory floor**	A real number between zero and one.	The capital ratio at which the bank is closed in the model (A3, A5).	No	Assumed by A5 at the lines the sources find binding, 8% for the baseline, and 10% to 2009Q4 and 10.5% from 2010Q1 for the enhanced model (M5).
R	**reference**	A real number, 0.105.	The ratio the facts are measured from.	No	Assumed, fixed in comparison/PREREG_comparison.md before any fact was computed.
μ_i	**mean earnings**	A real number.	The mean of the bank's yearly net income over the previous year's risk-weighted assets, less the riskless return on its capital ratio (A1).	No	Computed over ten years of December reports by the code of replication/peura_keppo_2006, each year as net income over risk-weighted assets less the riskless rate times the capital ratio (`EnhancementMap.net_earnings`).
σ_i	**earnings volatility**	A positive real number.	The standard deviation of the same yearly earnings (A6).	No	Computed over the same ten years from the same yearly terms (`EnhancementMap.net_earnings`).

Symbol	Name	Type	Represents	Directly observed	Obtained how
ρ	**discount rate**	A positive real number, 0.04.	The rate at which the bank discounts dividends, its cost of equity (A4).	No	Assumed at Peura and Keppo's value.
s_i	**judged risk**	A positive real number.	The volatility the buffer over the floor moves with in the model (A1, A6).	No	Assumed by A6 to be the earnings volatility or the book-capital volatility, or one of the measures of E3.
$\sigma_{D,i}$	**book-capital volatility**	A positive real number.	The standard deviation of the yearly change in the capital ratio over ten years (A6).	No	Computed by comparison/book_vol.py. The change is the same for the capital ratio and for the buffer over the floor (`EnhancementMap.change_free_of_floor`).
$\sigma_{E,i}$	**equity volatility**	A positive real number.	The annualised standard deviation of daily log changes of the parent's share price.	No	Computed from pinned daily prices by comparison/judged_risk.py as $\sqrt{252}$ times the daily standard deviation (`EnhancementMap.annualise`).
X_i	**mean capital ratio**	A real number above zero.	The mean of the bank's capital ratio over ten December reports.	No	Computed by comparison/judged_risk.py; a mean lies between the least and the largest ratio (`EnhancementMap.mean_between`).
$\sigma_{B,i}$	**market judged risk**	A positive real number.	The volatility of the capital ratio if book capital moved with the market value of equity (E3).	No	Computed as $\sigma_{E,i}X_i$, the standard deviation of X_i times a return (`EnhancementMap.market_scale`).
D	**stress-test decline**	A real number at least zero.	The fall in a holding company's tier 1 common ratio from its actual value to its projected minimum under the severely adverse scenario.	No	Computed from the published rows of lit/dfast_2013 and lit/dfast_2014 as the actual less the minimum (`StressTests.decline_of_rows`).
z	**normal quantile**	A positive real number.	The quantile of a normal loss that a decline is read as (M4).	No	Assumed on the grid 1.645, 2.326, 3.090, with 2.326 central.
$\sigma_{A,i}$	**stress-test judged risk**	A positive real number.	The volatility under which a nine-quarter decline D is the z-quantile (E3).	No	Computed as $D/(\frac{3}{2}z)$ (`StressTests.judged_sigma`).
$\sigma_{C,i}$	**own-loss judged risk**	A positive real number.	The same conversion applied to the bank's own worst nine-quarter decline (E3).	No	Computed by comparison/own_loss.py as $D/(\frac{3}{2}z)$ with the bank's decline (`StressTests.judged_sigma`).
$\sigma_{N,i}$	**do-nothing volatility**	A positive real number.	The standard deviation of the yearly change in Berger et al.'s do-nothing ratio (E8).	No	Computed by comparison/do_nothing.py; each change is an earnings part plus an asset part (`EnhancementMap.do_nothing_split`).
S_i	**bank size**	A real number.	The log of the bank's mean total assets (E9).	No	Computed from the Call Report field `ASSET`; the log preserves the order of sizes (`EnhancementMap.size_monotone`).

1.2 Internal objects

Table 2: Internal objects of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
x	**buffer over the floor**	A real number, the state.	The capital ratio less the regulatory floor (A1, A3).	No	Computed as $k - \underline{k}$ (`EnhancementMap.buffer_of_ratio`).
W_t	**Brownian motion**	A standard Brownian motion.	The source of the buffer's random movement (A1).	No	Never observed.
p	**positive root**	A positive real number.	The positive root of the value equation (M1).	No	Computed from μ_i, s_i and ρ (`PeuraKeppo.dplus_pos`).
m	**negative root**	A negative real number.	The negative root of the value equation (M1).	No	Computed from μ_i, s_i and ρ (`PeuraKeppo.dminus_neg`).
a_1, a_2	**barrier coefficients**	Two real numbers of opposite sign.	The weights of the two exponentials in the smooth-fit condition (M2).	No	A device of the proof of M2 (`PeuraKeppo.coeff_signs`).
u_0	**dividend barrier**	A positive real number.	The buffer over the floor at which dividends are paid (A2, M2).	No	Computed as $\frac{2}{p-m} \log \frac{-m}{p}$ (`PeuraKeppo.no_issue_barrier`).

1.3 Outputs

Table 3: Outputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
F_1	**thin-buffer dividend share**	A share between zero and one.	The share of quarters with dividends paid from a buffer over the reference below σ_i, observed or predicted.	No	Computed by comparison/facts.py and window.py on the data, and by the simulations on the model (`EnhancementMap.share_unit`).
F_2	**far-above share**	A share between zero and one.	The share of bank-quarters more than $4\sigma_i$ above the reference, observed or predicted.	No	Computed by the same scripts (`EnhancementMap.share_unit`).